

LECTURE 5: ARE MONOPOLISTS SOMETIMES GOOD?

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WEEK 2

ARTIFICIAL MONOPOLIES

We've been pretty hard on monopolies so far because they decrease welfare compared to a competitive market

But if we look around us, the government often uses policy to artificially create monopolies

Are there some circumstances where we would prefer to have a monopoly over competition?

DRUG DISCOVERY COSTS A LOT OF MONEY

Discovering new pharmaceutical costs a lot of money and is subject to many failures

- The average cost to develop a new pharmaceutical drug and get it to market between 2000-2018 was \$172 million
- There is about a 4:1 failure rate - for ever one drug that makes it to market, 4 fail
- This means that every drug needs to make **at least** \$878 million to turn a profit
- So why does any pharmaceutical business exist?

PHARMACEUTICALS ARE COVERED BY A PERIOD OF EXCLUSIVITY

- Depending on the type of drug, the developer gets the legal right to act like a monopoly for a certain period
- This provides the incentive for pharma companies to spend so much on new drug research
- Otherwise, other companies that haven't done the R&D work can essentially copy the drug for no upfront cost
- How do we think about this as policy makers?

- Orphan Drug Exclusivity (ODE) – 7 years
- New Chemical Entity Exclusivity (NCE) – 5 years
- Generating Antibiotic Incentives Now (GAIN) Exclusivity– 5 years added to certain exclusivities
- New Clinical Investigation Exclusivity – 3 years
- Pediatric Exclusivity (PED) – 6 months added to existing Patents/Exclusivity
- Patent Challenge (PC) – 180 days (this exclusivity is for ANDAs only)
- Competitive Generic Therapy (CGT) - 180 days (this exclusivity is for ANDAs only)

HOW DO WE THINK ABOUT THIS AS POLICY MAKERS?

- Typically we think that monopolies are bad, because they create deadweight loss in the economy
- But if it incentivises firms to make new products, then we should perhaps allow it
- If the pharma company would not make the investment for a new pharmaceutical in a competitive market, then we are better off allowing a monopoly
- If the pharma company **would** make the investment for a new pharmaceutical in a competitive market, then we should not allow the company to have monopoly power

HOW DO WE THINK ABOUT THIS AS POLICY MAKERS?

We would prefer a **competitive market** to a **monopoly**

We would prefer a **monopoly** to **no market at all**

EXAMPLE OF DRUG INVESTMENT

A pharmaceutical company makes an investment of \$100 million to get a new drug to market. If it does not expect to recover at least that much in producer surplus/profit, it will not make the investment.

The total cost of supplying the drug to market is $TC = Q + Q^2$

Demand for the drug is $Q_D = 10,000 - 0.1P$

If the company operates in a competitive market, it will capture 10% of the total producer surplus. If it operates in a monopoly, it will capture 100% of the surplus

Should we allow it to be a monopoly?

EXAMPLE OF DRUG INVESTMENT

First, let's find the company's profit in a competitive market

The marginal cost is $MC = \frac{dTC}{dQ} = 1 + 2Q$

The demand curve can be written as inverse demand: $P = 100,000 - 10Q_D$

In a competitive market, $P = MC$

$$1 + 2Q = 100,000 - 10Q$$

$$12Q = 99,999$$

$$Q = 8,333$$

So the total competitive market will sell 8,333 units of the drug

The price will be: $P = MC = 1 + 2Q = 16,667$

The profit of the total market will be:

$$\pi = P \times Q - TC = 16,667 \times 8,333 - (8,333 + 8,333^2) \approx \$69 \text{ million}$$

The original company will only get a 10% share of this profit of \$6.9 million, less than the original investment of \$100 million. So it will not create the drug

EXAMPLE OF DRUG INVESTMENT

Now let's find the company's profit if it has a period of exclusivity

The profit function is:

$$\begin{aligned}\pi &= P \times Q - (Q + Q^2) \\ &= P \times (10,000 - 0.1P) - (10,000 - 0.1P) - (10,000 - 0.1P)^2\end{aligned}$$

Find the price that maximizes profit:

$$\begin{aligned}\frac{d\pi}{dP} &= (10,000 - 0.1P) - 0.1P + 0.1 + 2 \times 0.1 \times (10,000 - 0.1P) = 0 \\ 10,000 + 0.1 + 2,000 - 0.1P - 0.1P - 0.02P &= 0 \\ -0.2P &= -12,000.1 \\ P &= 54,545\end{aligned}$$

So the monopoly price is much higher!

The pharma company will sell:

$$Q = 10,000 - 0.1P = 10,000 - 5,454 = 4,545$$

The pharma company's total profit is:

$$\begin{aligned}\pi &= 54,545 \times 4,545 - 4,545 - 4,545^2 \\ &= \sim \$227 \text{ million}\end{aligned}$$

So if the company gets exclusive rights, it earns enough profit to cover its investment

WHAT SHOULD WE TAKE AWAY FROM THIS?

From a policy perspective, there are sometimes when it's preferable to have a monopoly

- We should not want a monopoly if we could sustain a competitive market
- If our choice is between a monopoly and no market at all - we're still better off with a monopoly! (Total surplus under no market is always 0)

In general, for it to be better to have a monopoly, there needs to be sufficiently high costs of entry where to encourage the firm to enter

WHAT SHOULD WE TAKE AWAY FROM THIS?

- In the USA, it's the job of the Federal Trade Commission to analyze whether increasing market power in industries can be socially beneficial or not
- Sometimes mergers between firms can increase market power, but decrease the fixed costs of operating so that, on net, society is better off
- The Patent Office also exists to allow firms to claim exclusivity over novel inventions - ideally incentivizing innovation



OTHER POLICIES OUT THERE?

We don't always have to allow companies with large investment costs to act as monopolies

Other options exist and are used in parallel:

- Subsidising research costs
- Government guaranteeing minimum price and quantities (COVID vaccines)
- Nationalisation

NATURAL MONOPOLIES AND REGULATION

- The other place we might prefer monopolies is when it saves duplicated fixed costs
- If every competitor has to spend the same start up/fixed cost/investment cost, we might prefer to have just one monopoly
- We call these **natural monopolies**

NATURAL MONOPOLIES AND REGULATION

The electricity grid has an unusual cost profile:

- It has really, really high fixed costs to initially build the network
- But the marginal cost of serving customers once it's built is really low - ie, marginal costs are low

This is an example of a **natural monopoly** - to compete with the power grid, a competitor would have to waste billions of dollars replicating the entire grid

- It's socially beneficial to just operate one grid (saves the costs of building competitor grids)

NATURAL MONOPOLIES AND REGULATION

- The barriers of entry of a natural monopoly differ to that of a drug company
- With a natural monopoly, any competitor would have to duplicate the entire fixed cost of entering the market
 - Example: a competitor to PGE would have to build an entirely new electricity grid
 - Will be better for society to just have one grid to save the cost
- With the pharmaceutical industry - competitors could easily copy the new drug once the research has been done
 - Competitors don't have to duplicate the fixed cost so we might allow competition
 - Other example: Once someone has written a book (large fixed cost), it's very easy to copy it. We don't allow this under copyright law!

NATURAL MONOPOLIES AND REGULATION

When natural monopolies exist, the government is keen to regulate them:

- Power grids
- Railway networks
- Gas pipelines
- Internet backbone networks

We should think of regulation as useful when we could not sustain a competitive market because of high barriers to entry. There are a lot of jobs for government and consulting economists that regulate/fight regulation large companies

NATURAL MONOPOLIES AND REGULATION

How does the government regulate the electricity grid operator PGE? If it was not regulated, it would charge monopoly prices!

- In California, the California Public Utilities Commission sets a maximum price that PGE can charge
- The price is supposed to reflect the profit PGE would make if they invested their money into a more competitive infrastructure project
- It keeps the balance where PGE makes a profit (keeping PGE happy), but it is far below monopoly profit (keeping customers happy)
- CPUC also regulated service standards so PGE does not diminish its service quality

